

Module 1: Linguistic Foundations & Preprocessing

5 Linguistic Levels, BPE Tokenization, Stemming & Lemmatization

Module I: Linguistic Foundations & Preprocessing — Topics Covered

1. Five Levels of Linguistic Analysis
2. Ambiguity in Natural Language (4 Forms)
3. Tokenization & Byte-Pair Encoding (BPE)
4. Porter Stemmer vs. WordNet Lemmatization
5. Regular Expressions for NLP Information Extraction

1. Five Levels of Linguistic Analysis

Linguistic Level	Core Domain of Study	Example Task / Challenge
1. Phonetics / Phonology	Acoustic sounds, phonemes, and pronunciation rules.	Speech-to-Text acoustic modeling.
2. Morphology	Structure of words, morphemes, prefixes, suffixes, and inflections.	Stemming ('running' → 'run').
3. Syntax	Grammatical structure, phrase groupings, and word order.	POS Tagging and Parse Trees.
4. Semantics	Literal meaning of words, phrases, and sentence composition.	Word Sense Disambiguation (WSD).
5. Pragmatics / Discourse	Contextual meaning, speaker intent, coreference, and sarcasm.	Anaphora resolution ('Mary saw the car. She bought it.').